

Instituto Tecnológico de Costa Rica

Operations Research - Semester II

Simplex

Members:

Adrián Zamora Chavarría
Daniel Romero Murillo

Date: November 18, 2025

Dakota

Maximize

$$Z = 60.00x_1 + 35.00x_2 + 20.00x_3$$

Subject to

$$8.00x_1 + 6.00x_2 + 1.00x_3 \leq 48.00$$

$$4.00x_1 + 2.00x_2 + 1.50x_3 \leq 20.00$$

$$2.00x_1 + 1.50x_2 + 0.50x_3 \leq 8.00$$

$$0.00x_1 + 1.00x_2 + 0.00x_3 \leq 5.00$$

Simplex Table

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	-60	-35	-20	0	0	0	0	0
	0	8	6	1	1	0	0	0	48
	0	4	2	1.5	0	1	0	0	20
	0	2	1.5	0.5	0	0	1	0	8
	0	0	1	0	0	0	0	1	5

Intermediate Tables

Pivoting 1

Most Negative

Column 2 (-60.00)

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	-60	-35	-20	0	0	0	0	0
	0	8	6	1	1	0	0	0	48
	0	4	2	1.5	0	1	0	0	20
	0	2	1.5	0.5	0	0	1	0	8
	0	0	1	0	0	0	0	1	5

Fractions

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b	Frac
	1	-60	-35	-20	0	0	0	0	0	
	0	8	6	1	1	0	0	0	48	6.00
	0	4	2	1.5	0	1	0	0	20	5.00
	0	2	1.5	0.5	0	0	1	0	8	4.00
	0	0	1	0	0	0	0	1	5	inf

$$48.00/8.00 = 6.00$$

$$20.00/4.00 = 5.00$$

$$8.00/2.00 = 4.00$$

$$5.00/0.00 = \text{inf}$$

Smallest fraction: 4.00 $\rightarrow$ Pivot: row 4

Pivot

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	-60	-35	-20	0	0	0	0	0
	0	8	6	1	1	0	0	0	48
	0	4	2	1.5	0	1	0	0	20
	0	2	1.5	0.5	0	0	1	0	8
	0	0	1	0	0	0	0	1	5

Canonization

$$R_4 \leftarrow R_4/2.00$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	-60	-35	-20	0	0	0	0	0
	0	8	6	1	1	0	0	0	48
	0	4	2	1.5	0	1	0	0	20
	0	1	0.75	0.25	0	0	0.5	0	4
	0	0	1	0	0	0	0	1	5

$$R_1 \leftarrow R_1 + 60.00R_4$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	10	-5	0	0	30	0	240
	0	8	6	1	1	0	0	0	48
	0	4	2	1.5	0	1	0	0	20
	0	1	0.75	0.25	0	0	0.5	0	4
	0	0	1	0	0	0	0	1	5

$$R_2 \leftarrow R_2 + -8.00R_4$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	10	-5	0	0	30	0	240
	0	0	0	-1	1	0	-4	0	16
	0	4	2	1.5	0	1	0	0	20
	0	1	0.75	0.25	0	0	0.5	0	4
	0	0	1	0	0	0	0	1	5

$$R_3 \leftarrow R_3 + -4.00R_4$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	10	-5	0	0	30	0	240
	0	0	0	-1	1	0	-4	0	16
	0	0	-1	0.5	0	1	-2	0	4
	0	1	0.75	0.25	0	0	0.5	0	4
	0	0	1	0	0	0	0	1	5

$$R_5 \leftarrow R_5 + -0.00R_4$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	10	-5	0	0	30	0	240
	0	0	0	-1	1	0	-4	0	16
	0	0	-1	0.5	0	1	-2	0	4
	0	1	0.75	0.25	0	0	0.5	0	4
	0	0	1	0	0	0	0	1	5

Pivot Result

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	10	-5	0	0	30	0	240
	0	0	0	-1	1	0	-4	0	16
	0	0	-1	0.5	0	1	-2	0	4
	0	1	0.75	0.25	0	0	0.5	0	4
	0	0	1	0	0	0	0	1	5

Pivoting 2

Most Negative

Column 4 (-5.00)

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	10	-5	0	0	30	0	240
	0	0	0	-1	1	0	-4	0	16
	0	0	-1	0.5	0	1	-2	0	4
	0	1	0.75	0.25	0	0	0.5	0	4
	0	0	1	0	0	0	0	1	5

Fractions

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b	Frac
	1	0	10	-5	0	0	30	0	240	
	0	0	0	-1	1	0	-4	0	16	-16.00
	0	0	-1	0.5	0	1	-2	0	4	8.00
	0	1	0.75	0.25	0	0	0.5	0	4	16.00
	0	0	1	0	0	0	0	1	5	inf

$$16.00 / -1.00 = -16.00$$

$$4.00 / 0.50 = 8.00$$

$$4.00 / 0.25 = 16.00$$

$$5.00 / 0.00 = \text{inf}$$

Smallest fraction: 8.00 $\rightarrow$ Pivot: row 3

Pivot

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	10	-5	0	0	30	0	240
	0	0	0	-1	1	0	-4	0	16
	0	0	-1	0.5	0	1	-2	0	4
	0	1	0.75	0.25	0	0	0.5	0	4
	0	0	1	0	0	0	0	1	5

Canonization

$$R_3 \leftarrow R_3 / 0.50$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	10	-5	0	0	30	0	240
	0	0	0	-1	1	0	-4	0	16
	0	0	-2	1	0	2	-4	0	8
	0	1	0.75	0.25	0	0	0.5	0	4
	0	0	1	0	0	0	0	1	5

$$R_1 \leftarrow R_1 + 5.00R_3$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	0	0	-1	1	0	-4	0	16
	0	0	-2	1	0	2	-4	0	8
	0	1	0.75	0.25	0	0	0.5	0	4
	0	0	1	0	0	0	0	1	5

$$R_2 \leftarrow R_2 + 1.00R_3$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	0	-2	0	1	2	-8	0	24
	0	0	-2	1	0	2	-4	0	8
	0	1	0.75	0.25	0	0	0.5	0	4
	0	0	1	0	0	0	0	1	5

$$R_4 \leftarrow R_4 + -0.25R_3$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	0	-2	0	1	2	-8	0	24
	0	0	-2	1	0	2	-4	0	8
	0	1	1.25	0	0	-0.5	1.5	0	2
	0	0	1	0	0	0	0	1	5

$$R_5 \leftarrow R_5 + -0.00R_3$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	0	-2	0	1	2	-8	0	24
	0	0	-2	1	0	2	-4	0	8
	0	1	1.25	0	0	-0.5	1.5	0	2
	0	0	1	0	0	0	0	1	5

Pivot Result

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	0	-2	0	1	2	-8	0	24
	0	0	-2	1	0	2	-4	0	8
	0	1	1.25	0	0	-0.5	1.5	0	2
	0	0	1	0	0	0	0	1	5

Results

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	0	-2	0	1	2	-8	0	24
	0	0	-2	1	0	2	-4	0	8
	0	1	1.25	0	0	-0.5	1.5	0	2
	0	0	1	0	0	0	0	1	5

Objective Value

$$Z = 280.00$$

Variables

$$x_1 = 2.00$$

$$x_2 = 0.00$$

$$x_3 = 8.00$$

Slack or Surplus

$$s_1 = 24.00$$

$$s_2 = 0.00$$

$$s_3 = 0.00$$

$$s_4 = 5.00$$

Multiple Solutions

A non-basic variable has a 0 on its first row, allowing us to pivot again and find another optimal solution.

Pivoting 3

Most Negative

Column 3 (0.00)

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	0	-2	0	1	2	-8	0	24
	0	0	-2	1	0	2	-4	0	8
	0	1	1.25	0	0	-0.5	1.5	0	2
	0	0	1	0	0	0	0	1	5

Fractions

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b	Frac
	1	0	0	0	0	10	10	0	280	
	0	0	-2	0	1	2	-8	0	24	-12.00
	0	0	-2	1	0	2	-4	0	8	-4.00
	0	1	1.25	0	0	-0.5	1.5	0	2	1.60
	0	0	1	0	0	0	0	1	5	5.00

$$24.00 / -2.00 = -12.00$$

$$8.00 / -2.00 = -4.00$$

$$2.00 / 1.25 = 1.60$$

$$5.00 / 1.00 = 5.00$$

Smallest fraction: 1.60 $\rightarrow$ Pivot: row 4

Pivot

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	0	-2	0	1	2	-8	0	24
	0	0	-2	1	0	2	-4	0	8
	0	1	1.25	0	0	-0.5	1.5	0	2
	0	0	1	0	0	0	0	1	5

Canonization

$$R_4 \leftarrow R_4 / 1.25$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	0	-2	0	1	2	-8	0	24
	0	0	-2	1	0	2	-4	0	8
	0	0.8	1	0	0	-0.4	1.2	0	1.6
	0	0	1	0	0	0	0	1	5

$$R_1 \leftarrow R_1 + -0.00R_4$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	0	-2	0	1	2	-8	0	24
	0	0	-2	1	0	2	-4	0	8
	0	0.8	1	0	0	-0.4	1.2	0	1.6
	0	0	1	0	0	0	0	1	5

$$R_2 \leftarrow R_2 + 2.00R_4$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	1.6	0	0	1	1.2	-5.6	0	27.2
	0	0	-2	1	0	2	-4	0	8
	0	0.8	1	0	0	-0.4	1.2	0	1.6
	0	0	1	0	0	0	0	1	5

$$R_3 \leftarrow R_3 + 2.00R_4$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	1.6	0	0	1	1.2	-5.6	0	27.2
	0	1.6	0	1	0	1.2	-1.6	0	11.2
	0	0.8	1	0	0	-0.4	1.2	0	1.6
	0	0	1	0	0	0	0	1	5

$$R_5 \leftarrow R_5 + -1.00R_4$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	1.6	0	0	1	1.2	-5.6	0	27.2
	0	1.6	0	1	0	1.2	-1.6	0	11.2
	0	0.8	1	0	0	-0.4	1.2	0	1.6
	0	-0.8	0	0	0	0.4	-1.2	1	3.4

Pivot Result

	Z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
	1	0	0	0	0	10	10	0	280
	0	1.6	0	0	1	1.2	-5.6	0	27.2
	0	1.6	0	1	0	1.2	-1.6	0	11.2
	0	0.8	1	0	0	-0.4	1.2	0	1.6
	0	-0.8	0	0	0	0.4	-1.2	1	3.4